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Dingxian Cao

Quantitative Analyst/Researcher

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My Portfolio
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Highly trained Statistician and Econometrician with 5+ years of research experience, specialized in non-stationary time series(panel) structure, cooperating with interdisciplinary teams, developing most frontier big data models to identify true information and value. And delivering interpretation of techniques to the audience.

SKILLS

Tools and Languages	Python, Shell, Git, $\text{\LaTeX}$, Markdown, R, MySQL
Quantitative Research	Statistical Modeling, Economical Modeling, Machine learning, High-dimensional data Modeling
Communication	English, Chinese (native speaker)

WORKING EXPERIENCE

Quantitative Analytics and Model Dev Senior Analyst Jul 2022 — Present
PNC Bank, NA Tysons, VA

- Quantitative Model Risk Validation.

GA/RA Aug 2017 — May 2022
University of Connecticut Storrs, CT

- Lecturer: Econometrics I(2311), Python for Economists(3322).
- RA role: Prof.Kao, Prof. Choi, Department of Economics.
- TA: Econometrics I (6310), Econometrics III (6310).

RESEARCH PAPERS/PROJECTS

1. Kaggle competition: G-Research Crypto Forecasting Nov 2021 — Feb 2022
Rank: top 1% 18/1946

- Identify and forecast the non-stationary structure of cryptocurrency markets by *XGBoost*, *LSTM* and *High-dimensional Model*(PhD Thesis).
- Links: [Kaggle account](#), [Github showcase](#), [Empirical case PDF](#)

2. Inference on Panel AR Model with Possible Unit-roots Oct 2019 — Jan 2022
PhD Thesis

- Uniform Analysis of Dynamic Panel through Local-to-Unity. (submitted to *Econometrica*)
- Asymptotical analysis of persistent Panel model.
- Non-asymptotics approach of building inference on persistent Panel model.
- Non-parametric approach of modeling high-dimensional dependence structure.

3. Global Market Covariance Identification by Multilevel factor model Oct 2021 — Feb 2022
RA project

- Work as RA of Prof. SungHoon Choi to identify large covariance matrix and pricing factors of index component companies across global markets.

4. Non-asymptotics of High-dimensional Data 2018 — present
RA project

- Work as RA of Prof. Chihwa Kao to study non-asymptotical approaches in Econometrics.
- Field focus: *Concentration Inequalities of Measures*, *Wasserstein distance*, *MinMax Robustness*, *Reproducing kernel Hilbert space*.

EDUCATION

PhD in Economics , <i>University of Connecticut</i>	GPA: 4/4 Aug 2017 - May 2022
M.S. in Statistics , <i>University of Wisconsin-Madison</i>	GPA: 3.7/4 Aug 2015 - May 2017
B.S. in Statistics , <i>East China Normal University</i>	GPA: 3.5/4 Aug 2012 - May 2016

ACTIVITIES/AWARDS

New England Statistical Symposium 2020	May 2019
Timothy A. and Beverly C. Holt Economics Fellowship	Spring 2019
Graduate School Pre-Doctoral Fellowship	Spring 2021
G200 Youth Forum	May 2015
Cambridge Real Estate and Finance Seminar, CREFS	Jan 2013